

## Semester 1 · Academic Year 2026–2027

Family name: ..... Given name: .....

(1 mark)

[illegible]

| 1 |   |
|---|---|
| 0 | 1 |
|   |   |

(5 marks)

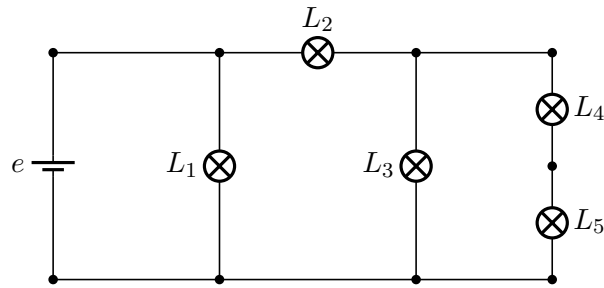
| ①                    |                      |                      |
|----------------------|----------------------|----------------------|
| 0                    | 1                    | 2                    |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |

| ②                    |                      |                      |
|----------------------|----------------------|----------------------|
| 0                    | 1                    | 2                    |
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| ③                    |                      |
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### Exercise 3. Reading a circuit diagram

(4 marks)



Base your answers on the diagram exactly as drawn, even though the same circuit could be drawn with fewer wires.

How many nodes?

- ☐ 2                      ☐ 3                      ☐ 4                      ☐ 9

How many branches?

- ☐ 4                      ☐ 5                      ☐ 6                      ☐ 11

How many two-terminal components, connecting wires included?

- ☐ 6                      ☐ 9                      ☐ 11                      ☐ 14

How many independent meshes (independent mesh equations)?

- ☐ 2                      ☐ 3                      ☐ 4                      ☐ 5

### Exercise 4. Energy received by a phone

(4 marks)

A phone is charged in the DC regime: the charger applies  $u = 5.0 \text{ V}$  and delivers  $i = 2.0 \text{ A}$  during  $\Delta t = 1 \text{ h } 30 \text{ min}$ . Compute the energy received by the phone, in joules, then in watt-hours.

|                     |                     |                                          |
|---------------------|---------------------|------------------------------------------|
| $u = 5.0 \text{ V}$ | $i = 2.0 \text{ A}$ | $\Delta t = 1 \text{ h } 30 \text{ min}$ |
|---------------------|---------------------|------------------------------------------|

[illegible]

### Marking

1

|                          |                          |
|--------------------------|--------------------------|
| 0                        | 1                        |
| <input type="checkbox"/> | <input type="checkbox"/> |

2

|                          |                          |
|--------------------------|--------------------------|
| 0                        | 1                        |
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3

|                          |                          |
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| 0                        | 1                        |
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4

|                          |                          |
|--------------------------|--------------------------|
| 0                        | 1                        |
| <input type="checkbox"/> | <input type="checkbox"/> |

— End of the test —